

## **9. Academic poster style**

# Purpose of posters

- Academic posters are becoming a common approach to communicating and presenting research. Traditionally this has been the pursuit of scientific disciplines;
- Posters act as a medium to advertise your work;
- They enable you to summarise your work – and to get your main points across to as many people as possible;
- Act as a conversation starter - they engage people in discussion about your work.

# Thinking about your poster

Key questions need to be answered before you get to the design stage. These are:

- Who is your audience?
  - Professionals, academics, the general public – the language used and messages given out need to meet the needs of your audience
- What is your hook?
  - Catchy titles provide the feature that may draw your audience in to peruse your poster. Is the message clearly stated and will it capture the attention of your audience?
- What is the purpose of your poster?
  - Consider what messages you wish to communicate and the audience receiving the messages. Remember you need to present a coherent snapshot of your research
- What are the guidelines for your poster?
  - Refer to your brief
- What is your message?
  - Effective posters deliver clear messages, content is highly visual

# Planning your poster

- An effective poster should be:
  - Legible from a distance of about two meters away,
  - Use a title that captures the attention of your audience,
  - Your title should be readable from about 3 - 4 meters away.
  - Use the right language for the right audience,
  - Avoid clipart wherever possible – due to pixel quality the resolutions may not suit your poster, images may blur.,
  - Have enough ‘white’ space – this aids readability,
  - Expect to work for a period of about five weeks on your poster!! This will be intermittently, so don’t panic but give yourself plenty of time to proof and revise accordingly.

# Things to think about

People read posters from left to right and top to bottom – consider your layout – portrait/landscape. Be guided by your conference brief.

Using too much jargon may confuse your audience, unless they are specialists in your field and already know the jargon.

What message will readers take home with them? – what will they remember about your work?

Colour schemes – think about the tones and hues, avoid large swathes of garish colours.

Word count – a typical poster will have between 300 – 500 words, audience dependent.

Font – Avoid using more than two different fonts. Choose a font such as Arial.

Use a good balance of text, pictures, charts and graphics. The text should support your images and vice versa.

The size of your poster and the orientation you will use.

Line spacing and text justification – left justification maybe easier to read.

Getting it printed – matt/glossy to laminate or not to laminate – a matt finish reduces glare. Laminated posters travel better so this might be worth considering – refer to conference brief.

# Title

Presenter details and affiliations

Introduction



Method



Image/table/graphic

Results



Image

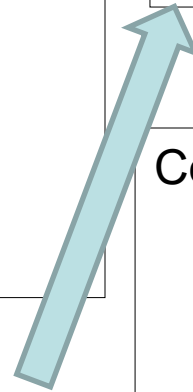
Image



Conclusion



References



# Quick Assessment of Basic Cognitive Function: A ‘Blood-Pressure Cuff’ for the Mind

Jennifer M. Shephard, Julia B. LeSage, Stephen M. Kosslyn

Harvard University, Department of Psychology, Cambridge, MA

Research supported by NASA cooperative agreement NCC 9-58 with the National Space Biomedical Research Institute

## Goals

- To design a quick and easily self-administered cognitive test battery, the *MiniCog Rapid Assessment Battery (MRAB)*, for immediate assessment of current cognitive functioning.
- To design software, *MiniCog* and *MiniCogWriter*, for scripting psychological tests and experiments to run on the Palm OS® and provide instantaneous feedback to user.

## Significance

- Addresses Critical Path Risks 19, 20, and 21 (human failure due to sleep/circadian rhythm disruption, interface/habitat/workload design problems, and neurobehavioral dysfunction); can be used
  - To assess effects of variables such as sleep loss on key cognitive and perceptual processes (CQ 6.08);
  - To assess effects of drugs, training, and other countermeasures on cognitive and perceptual processes;
  - For self-assessment by astronauts of their “current cognitive state” (CQ 6.11, 6.15, 6.20).
- Earth applications include use by truck drivers, surgeons, business travelers, air traffic controllers, mountain climbers, mission commanders, and others whose long hours, unusual shifts, or stressful environments may affect performance.

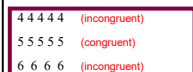
## Tests



**Perceptual Reaction Time** - Users must push the key corresponding to the location of a small oval on screen (squares represent different Palm screens).



**Vigilance (Attention)** - Users must respond with one key for random and infrequent targets and another key for distractors.



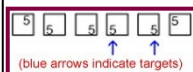
**Filtering (Attention)** - Users must indicate the number of digits on the screen; sometimes the number of digits and their meaning are congruent; on other trials they are not.



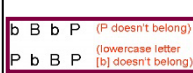
**Divided Attention** - Users must press one key for items of a particular shape or shade, and another key for items of another particular shape or shade.



**Verbal Working Memory** - Users must indicate which stimuli (presented one at a time, not in a row) are the same as the stimulus that appeared “two back.”



**Spatial Working Memory** - Users must indicate which stimuli are in the same location as the stimulus that appeared “two back” (boxes indicate separate Palm screens).



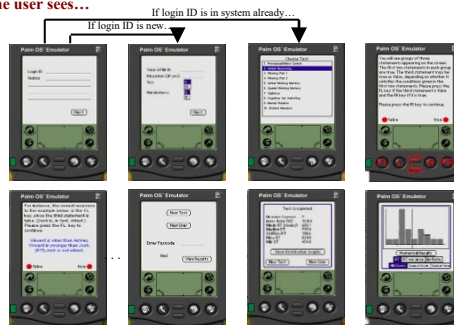
**Cognitive Set Switching** - Users must indicate which letter in each series of four does not belong; criterion for categorizing “odd man out” switches every 3 to 7 trials.



**Mental Rotation (Problem-solving)** - Users must decide whether two objects are the same or are mirror images.

**Verbal Reasoning (Problem-solving)** - Users must decide whether the third statement in series is true or false given information in the first two (see example on Palm screens in the center panel of this poster).

## What the user sees...



## Question

Does fatigue cause poorer performance (slower response times and/or more errors) selectively on the MRAB tests compared to baseline, and does caffeine ameliorate this effect?

## Participant characteristics

- Approximately 48 participants will be tested, equal numbers of males and females.
- Harvard students, between the ages of 18 and 30.
- Psychologically and physically healthy.
- Non-smokers.
- Regular consumers of caffeine who drink no more than 3 caffeinated beverages per day.

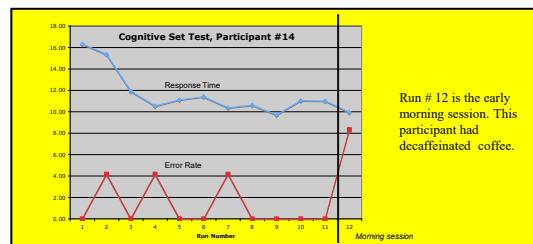
## Methods

- Participants practice MRAB tests a total of 10 to 12 times during four one-hour sessions between 9:00 AM and 3:00 PM, within a two-week period.
- At the end of the two weeks, participants are tested in a 5:00 AM session.
- All participants are given one cup of coffee (approx. 300 ml) prior to performing the early-morning tests; half receive caffeinated (approx. 150mg caffeine) coffee and half receive decaf (random and double-blind).
- Participants fill out questionnaires and other paperwork for 30 minutes after consuming coffee and prior to beginning MRAB.

## Results

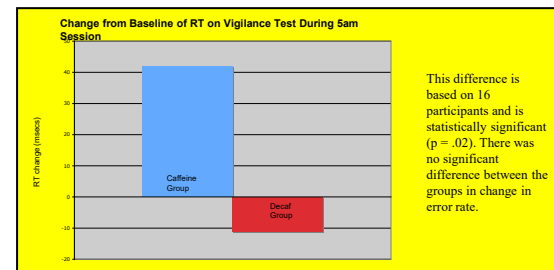
Data collection is still underway (approximately 22 participants have been tested so far), but data from a subset of the tested participants (16) is presented here.

Depending on the test, participants generally reach a baseline performance level after approximately 7 practice runs; see example performance curve below.



Because participants reach baseline after about 7 practice runs, we compared their performance during the early morning session with their average performance across the 3 practice runs prior to this session (corresponding to practice runs 8-10, 9-11, or 10-12, depending on the number of runs participants were able to complete during the practice sessions).

With the small number of participants included in these preliminary analyses, there were no statistically significant results except for the difference between the groups in response time performance on the Vigilance test; participants in the caffeine condition improved relative to baseline, while those in the decaf condition performed more poorly.



In addition, some trends were apparent; for instance, both groups appear to have slower Mental Rotation RTs during the early morning session, and faster Cognitive Set Switching and Perceptual Reaction Time RTs during this session -- but these results could be confounded by speed-accuracy tradeoffs.

Final analyses will include “throughput” as a performance measure (to take speed-accuracy tradeoffs into account) as well as response time variance. Participants’ subjective fatigue, objective hours of sleep prior to 5am session, and reported stress levels during all sessions will be included as covariates.

## Projects planned for upcoming year

- Continue validation of MRAB under stressful conditions
  - Comparison with other test batteries (e.g., WinSCAT [Wyle Laboratories]);
  - Use performance on MRAB to predict performance on an ecologically valid task (flight simulator) under stress;
  - Evaluate the effects of performance anxiety (e.g., public-speaking stress) on MRAB;
  - Continue collaborations with Dinges (sleep deprivation) and Lieberman (high-altitude and hockey-team head injury studies) laboratories.
- Extend MiniCog software to
  - Allow users to predict their performance just before and after completing a test (but prior to seeing their results);
  - Allow users to add notes or comments at the end of a test;
  - Include throughput as an automatically calculated performance measure;
  - Require two key-presses to exit the program (because users sometimes pressed an exit key by mistake);
  - Automatically present the next test on the list, instead of requiring users to choose (because they tend to forget which ones they’ve completed);
  - Include better randomization features for stimulus presentation.
- Continue discussions with collaborators about how to improve MiniCog user interface and possible additional tests to include.



# Cooling Effects of Dirt Purge Holes on the Tips of Gas Turbine Blades



Eric Couch, Jesse Christophel, Erik Hohlfeld, and Karen Thole



Gas turbine engines run better at higher combustion temperatures

At higher combustion temperatures, these engines generate more power and use less fuel. However, these temperatures are restricted by melting temperatures of the turbine blades downstream of the combustor (see Figure 1).

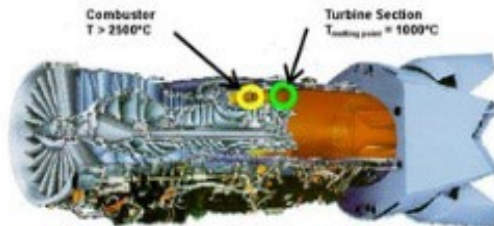


Figure 1. Pratt & Whitney F119 gas turbine engine.

Dirt purge holes on turbine blade tips allow for higher combustion temperatures

Harmful hot gases from the combustor leak across the gap between the blade tip and the shroud (see Figure 2). Dirt purge holes expel foreign particles from the blade tip so that film cooling holes are not blocked.

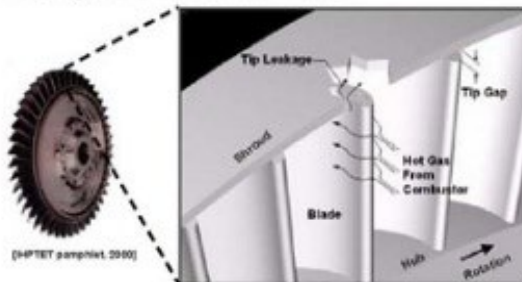


Figure 2. Flow at the tip region of a turbine blade.

The project goal was to find the film cooling effects of these dirt purge holes

To find the effects, we performed wind tunnel experiments with scaled turbine blades. The wind tunnel was low speed and low temperature, and the blades, shown in Figure 3, were scaled at 12 times their normal size. To measure temperatures on the blade tip, we used an infrared camera. Tip gap sizes and amount of coolant flow from the dirt purge holes were both varied.



Figure 3. Large-scale turbine blade in wind tunnel.

Temperature measurements were converted to dimensionless cooling effectiveness

$$\text{Effectiveness } \eta = \frac{T_{\infty} - T_{aw}}{T_{\infty} - T_c} \quad \text{where } \begin{matrix} T_{\infty} = \text{mainstream temperature} \\ T_{aw} = \text{constant temperature} \\ T_c = \text{adiabatic wall temperature (on tip surface)} \end{matrix}$$

Cooling increased with blowing ratio

The effectiveness contours of Figure 4 show that cooling increased with blowing ratio.

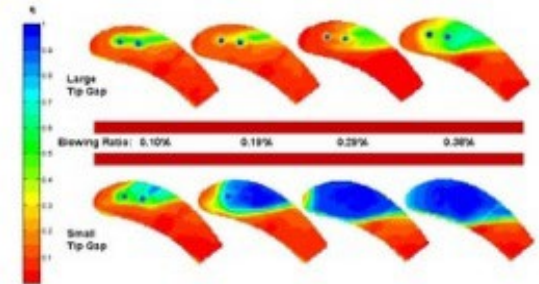


Figure 4. Measurements of film cooling effectiveness.

Tip size dramatically affected cooling

In Figure 5, the lateral averages of effectiveness plotted against the axial chord length show that tip size dramatically affected the cooling.

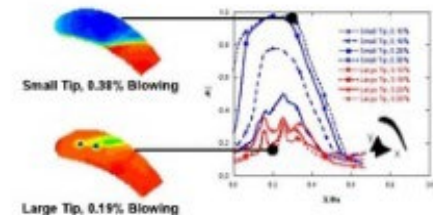


Figure 5. Laterally averaged effectiveness plotted against normalized axial chord.

In summary, dirt purge holes provide cooling to the tip surface

While intended to remove dirt from the blade, dirt purge holes also provide cooling to the tip surface. This cooling is enhanced with a small tip gap as the dirt purge floods the tip region near the leading edge with cool air.

Acknowledgments

The sponsor for this project was Pratt & Whitney.



# Creative Uses of Custom Electronics for Environmental Monitoring at the Christina River Basin CZO

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## Introduction

The Christina River Basin CZO is currently operating dozens of homemade dataloggers that are connected to different types of environmental sensors, and all the loggers were built using Arduino-based microprocessor boards. We have also created a variety of helpful devices to perform functions that would have been too expensive or impossible to implement with standard methods and commercial hardware. The ability to design and build custom electronic devices specifically suited to a unique task has gotten easier and cheaper, thanks to the recent popularity of open source electronics platforms like Arduino.



Arduino boards are available in a variety of sizes and configurations. Some have additional features like Xbee radio sockets, Ethernet ports and memory card slots. All of the boards shown here contain the same type of processor and are programmed the same, but the smaller boards are easier to fit into finished projects. Our dataloggers use a custom sensor interface board to easily connect various devices to the main board.



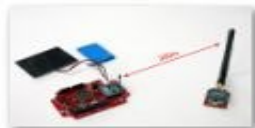
An interface board with memory card and clock creates a basic datalogger.



Xbee radio modules in the same network can use different power levels and configurations.



The Xbee radios form a self-healing mesh network to transmit the data back to the coordinator.



A solar powered logger with Xbee has a range of 350m to the base station, or 100m between logger nodes.



Some of the commercial sensors that can easily connect to Arduino loggers: soil moisture, conductivity, redox, CO<sub>2</sub> probes, pressure transducers, and oxygen sensors.

## Observations

- Temperature extremes cause batteries to fail or have significantly shorter lifespans. Live data networks help with logger maintenance.
- Solar panels mounted inside a clear enclosure cause excessive heat buildup on sunny days in the summer. External solar panels cost more but keep the circuitry and battery cooler.
- Humidity and condensation inside the enclosures can cause board failure. A conformal coating should be applied to the circuit boards before deployment.
- Xbee radio range is highly dependent on terrain, vegetation, and antenna gain. Sleep routines are needed to preserve battery life.
- Management of the live streaming data is a very complicated process.
- Our loggers are currently collecting over 30,000 datapoints each day.

## Practical Implementations of Arduino Hardware



**6-channel soil moisture probe logger**  
A quick and inexpensive (\$60) way to measure soil conditions.



**Power Usage Monitor and Logger**  
Measures power consumption by using non-invasive current transformers. It can be used on single circuits or a whole house. Also measures air temperature and humidity and stores all data to a memory card.



**Pressure Transducer Readout & Portable Soil Moisture Probe Tester**  
Useful for testing probes during deployment.



**Radio Reporting Streamgauge**  
An Arduino logger with Xbee radio sends live stream depth data to a base station that posts it to a web page.



**Solar powered well depth logger**  
Measures water depth in 4 wells, stores data to memory card and transmits via Xbee mesh network.



**Educational Sensor Integration Kit**  
A custom PC board allows students to easily experiment with a variety of basic sensors and then display the data on an LCD or transmit it via Xbee radio to a PC or other logger. The collected data can be submitted to an interactive data visualization system that maps and graphs the students' data alongside live streaming data from scientific stations operated by the USGS, NOAA, CZO, and others.



**Portable 3-chamber Soil Respirometer**  
The Arduino circuit displays live data from a LI-COR CO<sub>2</sub> analyzer, logs the data, and operates solenoid valves and pumps to control the flow of air through custom respiration chambers.



**60-chamber Respirometer Controller**  
Similar to the portable unit (left), this Arduino logger controls a LI-COR analyzer, pumps, and 120 separate solenoid valves that sample the rate of respiration in 60 jars of soil samples.



**Remote Streamwater Sampling System**  
(Also known as Dial-a-Pump) Researchers can remotely trigger a pump by calling the prepaid phone and then following the interactive voice menus to select pump options and hear live sensor value readouts. The pump system can also be adapted to be controlled via the internet or Xbee radio network.



The Dial-a-Pump controller operates a bank of large solenoid valves that can fill multiple barrels for sampling the stream water during different parts of a storm.

For more information, visit the CRB-CZO Arduino Project webpage where we will share details of our projects, schematics, and source code: <http://www.stroudcenter.org/research/projects/czo/arduino.shtml>



# Mobile Applications for eTourism

## Assumptions

- \* MDA as a personal device
- \* capable mobile devices
  - OS, GUI-Library, WebServices etc.
  - access to internet via GPRS, WLAN or UMTS
  - good graphical displays (320 x 280)
- \* GPS receiver (precision ca. 1m in 2007)
- \* Headset connected via. Bluetooth
  - ➔ future trends for tourism
    - > mobile agent as a tour guide
    - > location-based Multi-Player Games



## Dynamic Tour Guide

- \* DTG knows my personal interests
- \* What can I do in 3 hours?
- ➔ plan and offer an individual context aware tour (interests, position, etc.)
- ➔ navigation to the offered sights
- ➔ additional information about the sights (via. Audio)
- ➔ tour adaption (walking speed, sightseeing, breaks etc.)

## Scenario

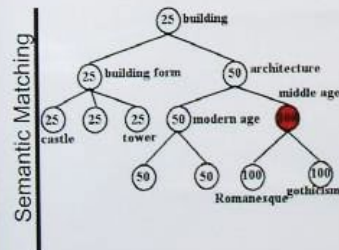
### Multi-Player-Games

- \* Monopoly in a real city ➔ GoNoPoly
- \* constantly checks the position
- \* playful discovery of a city (buy a street and collect rents)
- \* getting into contact with other players (event card)

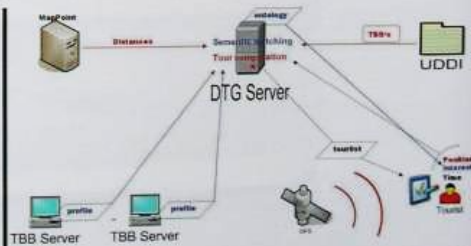


## Challenges

- \* gathering of the tourists interests
- \* reuse the interests of the tourist (interest profile)
- \* discovery of all sights in the tourists environment
- \* evaluation and selection of the sights (Semantic Matching)
- \* tour computation and navigation
- \* tour modification (automatically, manually)



## Architecture



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